



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Assisted Python Programming for Finance

TGS Ref No: TGS-2025052659

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 31.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
30.0	2 May 2026	Previous release — IBF-accredited 4-day lesson plan for AI Assisted Python Programming for Finance.	Dr. Alfred Ang
31.0	20 July 2026	Rebuilt from the single-source content module. Expanded to 62 hands-on finance labs across the nine topics; all labs migrated to the uv project manager; added a progressive Streamlit finance-dashboard capstone in Topics 7 and 9; full build-out of Topics 5-9 (exception handling, pandas, aggregation/visualisation, OOP, apply/pipe).	Dr. Alfred Ang

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Course Information

Course Title	AI Assisted Python Programming for Finance
Course Reference	TGS-2025052659
Accreditation	Skills Framework for Financial Services · IBF Standard FSE-DIT-3018-1.1 · Funded under IBF-STs
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	4 days · 8 training hours per day (32 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on AI-assisted Python labs using uv, pandas, yfinance and Streamlit
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Translate business requirements in financial services into Python programming objectives, and set up an AI-assisted Python development environment.
- LO2: Apply Python data types, operators and control structures to solve financial calculation and classification problems.
- LO3: Construct reusable functions and lambda expressions to compute financial ratios and metrics for business use cases.
- LO4: Implement error handling to build robust financial programs that survive missing, zero or malformed market data.
- LO5: Import, clean, filter and process financial data using the pandas package.
- LO6: Aggregate and visualise financial data with groupby, pivot tables and charts, and publish the results as an interactive Streamlit application.
- LO7: Apply object-oriented programming to model financial instruments and analysers.
- LO8: Analyse financial data using apply and pipe pipelines, assess code to identify gaps, and test the results.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), open-ended, 1 hour, open book.
- Practical Test (PP) — 7 open-ended practical tasks based on the Financial Data Solutions Inc. scenario, mapped to abilities A1.1–A1.9, 120 minutes, open book.
- Format: Open Book — course slides, Learner Guide and lab files only.
- Passing score: 80%.
- Final assessment is conducted on Day 4 from 5:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and IBF-STs funding. Open book assessment includes the slides, Learner Guide and lab files only.

Course Schedule

Day 1 — Python Foundations for Finance — environment, data types and control structures

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, learning outcomes, ground rules and mandatory digital attendance (AM)
10:00–10:30	30 min	Topic 1 — Introduction to Python Programming: Business requirements · Applications in financial services · AI-assisted Python IDE setup with uv (concepts + demonstration)
10:30–10:45	15 min	Tea break
10:45–11:30	45 min	Hands-on: Lab 1: Set Up an AI-Assisted Python Environment with uv; Lab 2: Hello World — Your First Finance Script; Lab 3: CPF Contribution Calculator; Lab 4: From Business Requirement to Programming Objective — SGX Price Fetcher
11:30–13:00	90 min	Topic 2 — Data Types and Operators (concepts). Hands-on: Lab 5: Number and String Literal Formatting for Financial Reporting; Lab 6: Loan Amortisation Calculator; Lab 7: Retirement Savings Calculator; Lab 8: String Methods and Extracting the Company Name from an Email
13:00–14:00	60 min	Lunch break
14:00–15:45	105 min	Digital attendance (PM). Hands-on: Lab 9: Lists — Top 10 SGX Companies by Market Capitalisation; Lab 10: Tuples — Immutable Instrument Identifiers; Lab 11: Dictionaries — Company Stock Values and Market Cap Lookup; Lab 12: Sets and Operators — Common Holdings Across Portfolios
15:45–16:00	15 min	Tea break
16:00–18:15	135 min	Topic 3 — Problem Solving with Control Structures (concepts). Hands-on: Lab 13: If-Else PE Ratio Valuation — BUY / HOLD / SELL; Lab 14: Activity: Return on Equity (ROE) Screener with a BUY /

		SELL Threshold; Lab 15: If-Elif Income Grouping and Credit Risk Banding; Lab 16: Ternary Operator — One-Line Trading Decisions; Lab 17: While Loop — Fibonacci Sequence and the Golden Ratio in Technical Analysis; Lab 18: For Loop and Range — Top 10 SGX Return on Assets (ROA) Ranking; Lab 19: Enumerate, Zip, Break and Continue — Combined ROE and ROA Scanner; Lab 20: Comprehensions — Net Profit Margin for 10 Singapore Stocks in One Line
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Reusable and Robust Code — functions, lambda and exception handling

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap, Q&A and mandatory digital attendance (AM)
9:45–10:45	60 min	Topic 4 — Scripting with Function and Lambda: Functions · Return values · Argument styles · Lambda · Map · Filter (concepts + demonstration)
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Hands-on: Lab 21: Functions — A Reusable Financial Formula Library; Lab 22: Return Multiple Values — A Full Income Statement Analysis in One Call; Lab 23: Activity: Net Profit Margin Function with Zero-Revenue Handling; Lab 24: Multiple, Default and Named Arguments — A Flexible Valuation Toolkit
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	Digital attendance (PM). Hands-on: Lab 25: Variable Arguments — *args and **kwargs for Portfolios of Any Size; Lab 26: Activity: Max ROE Function — Finding the Best Performer in a Portfolio; Lab 27: Lambda and Map — Gross Profit Margin and ROE Across a Portfolio; Lab 28: Filter — Top Income Customers and High-ROE

		Stock Screening
15:00–15:45	45 min	Topic 5 — Error Handling Using Exception: Exceptions versus syntax errors · Try and Except · Else clause · Finally (concepts + demonstration)
15:45–16:00	15 min	Tea break
16:00–18:15	135 min	Hands-on: Lab 29: Exceptions vs Syntax Errors in a Price Loader; Lab 30: Try/Except on a Failing Market-Data Fetch; Lab 31: ZeroDivisionError in an ROE and Ratio Calculator; Lab 32: KeyError and ValueError in a Portfolio Input Handler; Lab 33: Else, Finally and a Custom InsufficientDataError for a Risk Model; Lab 34: Activity: Loan Risk Classifier with Guarded Data Loading
18:15–18:30	15 min	Day 2 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Financial Data Analysis — pandas, aggregation, visualisation and Streamlit

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap, Q&A and mandatory digital attendance (AM)
9:45–10:45	60 min	Topic 6 — Import and Process Finance Data: Pandas package · DataFrame and Series · Import · Filter and slice · Clean missing data (concepts + demonstration)
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Hands-on: Lab 35: Pandas Series vs DataFrame with Price Data; Lab 36: Import Finance Data from CSV; Lab 37: Import Live Market Data with yfinance; Lab 38: Inspect a Dataset with head, info and describe
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	Digital attendance (PM). Hands-on: Lab 39: Filter and Slice by Date, Sector and Ticker; Lab 40: Handle Missing Market Data and Compare the Choices; Lab 41: Derived Columns — Daily Returns and Moving Averages; Lab 42: Activity: Algorithmic Trading — Moving Average Crossover

		Backtest
15:00–15:45	45 min	Topic 7 — Aggregate and Visualize Finance Data: Concat, append and merge · Groupby · Pivot table · Assess code for gaps · Test and visualise · Streamlit (concepts + demonstration)
15:45–16:00	15 min	Tea break
16:00–18:15	135 min	Hands-on: Lab 43: Generate Mock Credit-Loan Data with <code>df_finance</code> ; Lab 44: Join Finance Datasets with <code>concat</code> , <code>append</code> and <code>merge</code> ; Lab 45: Groupby: Annual Income by Gender and Region; Lab 46: Loan Application Counts by Status and Type; Lab 47: Multi-Metric Aggregation by Employment Sector with <code>.agg()</code> ; Lab 48: Pivot Table of Debt-to-Income Ratio and Matplotlib Visualisation; Lab 49: Streamlit Capstone Part 1 — Portfolio & Loan Analytics App Shell; Lab 50: Streamlit Capstone Part 2 — Charts and Pivot Tab in the Dashboard
18:15–18:30	15 min	Day 3 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 4 — Object-Oriented Finance and Analysis — OOP, apply/pipe, capstone and assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap, Q&A and mandatory digital attendance (AM)
9:45–10:45	60 min	Topic 8 — Object Oriented Programming: Classes and objects · Methods and overloading · Initializer and destructor · Inheritance · Polymorphism (concepts + demonstration)
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Hands-on: Lab 51: Build a Stock Class — Attributes, Methods and <code>__init__</code> ; Lab 52: A Portfolio Class Holding Many Stock Objects; Lab 53: Dunder Methods — <code>__str__</code> , <code>__repr__</code> , <code>__len__</code> and <code>__add__</code> to Combine Portfolios; Lab 54: Inheritance — an Instrument Base Class with Equity and Bond Subclasses; Lab 55:

		Polymorphism — One .value() Call Across a Mixed Instrument Book; Lab 56: StockTechnicalAnalyzer — an OOP Class Computing SMA, RSI and BUY/SELL Signals
13:00–14:00	60 min	Lunch break
14:00–14:45	45 min	Topic 9 — Analyze Finance Data: Improve code with apply and pipe · Applications of statistics · Analyse finance data to track changes (concepts + demonstration). Digital attendance (PM)
14:45–15:45	60 min	Hands-on: Lab 57: Credit Risk Classification with .apply(); Lab 58: Lambda versus Named Functions in .apply(); Lab 59: Building a clean → enrich → score Pipeline with .pipe(); Lab 60: Descriptive Statistics, Correlation and Tracking Change Over Time; Lab 61: Streamlit Capstone Part 3 — Technical Analysis and Risk Scoring Pages; Lab 62: Streamlit Capstone Finale — Test, Polish and Deploy the Analytics App
15:45–16:00	15 min	Tea break
16:00–16:30	30 min	Course recap, Q&A, Course Feedback and TRAQOM survey
16:30–16:45	15 min	Briefing for Assessment. Assessment digital attendance
16:45–17:45	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), open-ended, 1 hour, open book.
17:45–19:45	120 min	Practical Test (PP) — 7 open-ended practical questions based on the Financial Data Solutions Inc. scenario, 120 minutes, open book

Total training time: 555 minutes (9 hours 15 min) — 360 minutes instruction plus 195 minutes of assessment.

Lab Reference (aligned to the course topics)

Topic	Weighting	Labs
Topic 01: Introduction to Python Programming	8%	Lab 1, Lab 2, Lab 3, Lab 4
Topic 02: Data Types and Operators	12%	Lab 5, Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12
Topic 03: Problem Solving	14%	Lab 13, Lab 14, Lab 15, Lab

with Control Structures		16, Lab 17, Lab 18, Lab 19, Lab 20
Topic 04: Scripting with Function and Lambda	14%	Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27, Lab 28
Topic 05: Error Handling Using Exception	10%	Lab 29, Lab 30, Lab 31, Lab 32, Lab 33, Lab 34
Topic 06: Import and Process Finance Data	14%	Lab 35, Lab 36, Lab 37, Lab 38, Lab 39, Lab 40, Lab 41, Lab 42
Topic 07: Aggregate and Visualize Finance Data	14%	Lab 43, Lab 44, Lab 45, Lab 46, Lab 47, Lab 48, Lab 49, Lab 50
Topic 08: Object Oriented Programming	8%	Lab 51, Lab 52, Lab 53, Lab 54, Lab 55, Lab 56
Topic 09: Analyze Finance Data	6%	Lab 57, Lab 58, Lab 59, Lab 60, Lab 61, Lab 62